

Diffusion Equation Method of Global Minimization: Performance for Standard Test Functions¹

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Abstract. Recently, we have proposed a method of global minimization that is based on solving the diffusion equation with the objective function as a boundary condition. In the present paper, the performance of the method is examined when applied to the standard test functions of the Goldstein-Price, Hartman, Shekel, and Griewank families. It turns out that the method succeeds in all these cases. A comparison of the effectiveness of our method and other methods is also reported.

Key Words. Global optimization, global minimum, diffusion, multiple minima.

1. Introduction

Global minimization of a multivariate function represents a field of real challenge. The importance of the problem is obvious and comes from practically all domains of science and technology. As stated by Dixon *et al.* in 1975 (Ref. 1), "the present situation with regard to solving global optimization problems is still very poor." After 13 years, despite considerable progress achieved, the problem has not been solved and one needs more than ever a reliable global optimization method.

On the other hand, it has been proved that it is impossible to construct an algorithm that would be able to locate the global minimum of any function (Ref. 2). Hopefully, the functions encountered in physics are less pathological, because there the value of a function in an infinitesimal domain

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